Analytical Mechanics

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1 Brachistochrone's Curve

We will give the analytical solution to the Brachistochrone Problem proposed by Bernoulli and solved by Euler.

Use the cartesian plane, and set the starting position to be $(0, h)$ and the ending position to be (a, b), $0 \leq b \leq h$. Make the assumptions that

1. The ball is always in contact with the curve
2. The ball is of negligible size
3. Friction is negligible

Furthermore, let the curve be defined as a function $y(x)$.

Lemma (Fundamental Theorem of the Calculus of Variations). Let f be a fixed function in x, and let g be an arbitrary function in x. Then

$$\int_{x_0}^{x_1} f(x)g(x)\ \mathrm{d}x = 0$$

implies that f is identically 0.

Proof. Suppose for the sake of contradiction that f is not 0, so $\exists [a, b] \subseteq [x_0, x_1]$: $f(k) \neq 0, k \in [a, b]$. It can be shown that there exists a smooth function h satisfying $h(a) = h(b) = 0$ such that

$$\int_{x_0}^{x_1} f(x)h(x) \neq 0$$

so just add h to g to obtain a contradiction. □

Make the observations that

$$m\frac{d^2s}{dt^2} = mg\sin\theta = mg\frac{dy}{ds}(t)$$

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \iff \frac{ds}{dt} = \frac{dx}{dt}\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Combine both to get

$$\frac{1}{2}\frac{d}{dt}\left(\frac{ds}{dt}\right)^2 = g\frac{dy}{dt} \iff \frac{ds}{dt} = \sqrt{2g(h - y)}$$

Remark. A quicker way to derive $\frac{ds}{dt}$ is

$$\frac{1}{2}mv^2 = mg\Delta h = mg(h-y) \implies \frac{ds}{dt} = v = \sqrt{2g(h-y)}$$

but second derivatives are more fitting for this derivation.

To get time, we integrate

$$\int_0^a \frac{\sqrt{1+\left(\frac{dy}{dx}\right)^2}}{\sqrt{2g(h-y)}}\,\mathrm{d}x = \frac{1}{\sqrt{2g}}\int_0^a \sqrt{\frac{1+\left(\frac{dy}{dx}\right)^2}{h-y}}\,\mathrm{d}x$$

Let $F(y,y') = \sqrt{\frac{1+\left(\frac{dy}{dx}\right)^2}{h-y}}$ and define $y(x) = y_m(x) + \alpha\epsilon(x)$ for some function ϵ defined in x. Further define

$$I(F) = \frac{1}{\sqrt{2g}}\int_0^a F(y,y')\,\mathrm{d}x$$

as a function $I : F \to \mathbb{R}$.

Remark. We now eliminated t and x from the calculations. All that remains is to optimize y and y'. It suffices to treat these as separate functions.

Lemma (Euler-Lagrange).

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\frac{\partial F}{\partial y'} = 0$$

Proof.

$$\frac{dI}{d\alpha} = \frac{1}{\sqrt{2g}}\int_0^a \frac{\partial F}{\partial y}\cdot e(x) + \frac{\partial F}{\partial y'}\cdot\frac{d\epsilon}{dx}\,\mathrm{d}x$$

but $\epsilon(0) = \epsilon(a) = 0$, so

$$\int_0^a \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\epsilon(x)\right)\,\mathrm{d}x = 0$$

$$\implies \frac{dI}{d\alpha} = \frac{1}{\sqrt{2g}}\int_0^a \frac{\partial F}{\partial y}\epsilon(x) - \frac{d}{dx}\frac{\partial F}{\partial y'}\epsilon(x)\,\mathrm{d}x$$

$$\implies \frac{\partial F}{\partial y} - \frac{d}{dx}\frac{\partial F}{\partial y'} = 0$$

Remark. This is just a natural step in the proof. Only calling it a lemma because I need to cite it in the next lemma.

□

Lemma (Beltrami Identity).

$$\frac{d}{dx}\left(F - \frac{\partial F}{\partial y'}y'\right) = 0$$

$$\implies F - \frac{\partial F}{\partial y'}y' = C$$

for some constant C.

Proof.

$$\frac{d}{dx}F = \frac{\partial F}{\partial y}\cdot\frac{\partial y}{\partial x} + \frac{\partial F}{\partial y'}\cdot\frac{\partial y'}{\partial x}$$

because $\frac{\partial F}{\partial x} = 0$. Then

$$\frac{d}{dx}\left(\frac{\partial F}{\partial y'}y'\right) = \frac{\partial F}{\partial y'}\cdot\frac{\partial y'}{\partial x} + \frac{\partial F}{\partial y}y'$$

by Euler-Lagrange. Subtract to get

$$\frac{d}{dx}\left(F - \frac{\partial F}{\partial y'}y'\right) = 0$$

□

Note that

$$\frac{\partial F}{\partial y'} = y'\frac{1}{\sqrt{h-y}\sqrt{1+y'^2}}$$

and by Beltrami,

$$\sqrt{\frac{1+y'^2}{h-y}} - y'^2\frac{1}{\sqrt{h-y}\sqrt{1+y'^2}} = C$$
$$\implies \frac{1}{C} = \sqrt{h-y}\sqrt{1+y'^2}$$

is constant. Note that $C \neq 0$, because otherwise we have no solutions in y, y'.

Suppose that y' is strictly positive or strictly negative on the interval $[p, q]$. Substitute

$$\frac{dy}{dx} = \pm\sqrt{\frac{1-C^2(h-y)}{C^2(h-y)}}$$

to get

$$x = \int\frac{1}{y'}\,\mathrm{d}y = \int\pm\sqrt{\frac{C^2(h-y)}{1-C^2(h-y)}}\,\mathrm{d}y$$

Substitute $h - y = \frac{1}{C^2}\sin^2\theta$ gives $\frac{dy}{d\theta} = -\frac{2\sin\theta\cos\theta}{C^2}$ so

$$x = \int\pm\frac{\sin\theta}{\cos\theta}\cdot\frac{-2\sin\theta\cos\theta}{C^2}\,\mathrm{d}\theta = \mp\frac{1}{C^2}(\theta - \sin\theta\cos\theta)$$

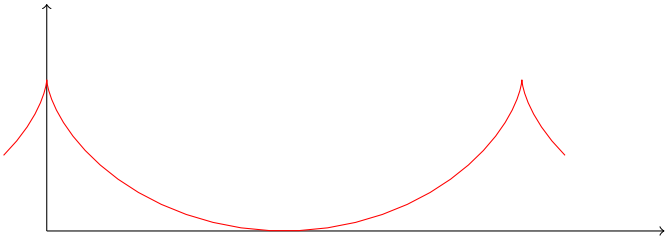
for certain bounds on θ. The parameterization is

$$\begin{cases} x = \frac{1}{C^2}(\theta - \sin\theta\cos\theta) \\ y = -\frac{1}{C^2}\sin^2\theta + h \end{cases}$$

where

$$\frac{a}{b-h} = \frac{\theta - \sin\theta\cos\theta}{\sin^2\theta}$$

can be used to solve for θ and C.



The Cycloid

2 Catenaries

We will an analytical solution to the hanging chain problem. The elementary solution can be found at the end of the document. Suppose that a string ℓ is attached to two points $(0, h)$ and (a, b). Make the assumptions that

1. The string is perfectly incompressible
2. The string has a uniform mass density

Furthermore, let the curve be defined as a function $y(x)$. We will minimize the potential energy

$$I = \int_0^a y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\ \mathrm{d}y$$

Let $F(y, y') = y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ and define $y(x) = y_m(x) + \alpha\epsilon(x)$. As usual, set

$$\frac{dI}{d\alpha} = \int_0^a \frac{\partial F}{\partial y}\epsilon(x) + \frac{\partial F}{\partial y'}\frac{d\epsilon}{dx}\ \mathrm{d}x = \int_0^a \frac{\partial F}{\partial y}\epsilon(x) - \frac{d}{dx}\frac{\partial F}{\partial y'}\epsilon(x)\ \mathrm{d}x = 0$$

and

$$F - \frac{\partial F}{\partial y'}y' = C$$

by Beltrami. Direct computation gives

$$y\sqrt{1 + y'^2} - yy'^2 \cdot \frac{1}{\sqrt{1 + y'^2}} = C \implies C = \frac{y}{\sqrt{1 + y'^2}}$$

Suppose that y' is strictly increasing or decreasing over an interval $[p, q]$. Substitute $\frac{dy}{dx} = \pm\sqrt{\frac{y^2 - C^2}{C^2}}$ so given correct bounds of integration,

$$x = \int \frac{1}{y'}\ \mathrm{d}y = \int \pm\sqrt{\frac{C^2}{y^2 - C^2}}\ \mathrm{d}y$$

which we may solve by substituting $y = C\sec\theta$ and $\frac{dy}{d\theta} = C\sec\theta\tan\theta$ to get

$$x = C\int \frac{\cos\theta}{\sqrt{C^2 - C^2\cos^2\theta}} C\sec\theta\tan\theta\ \mathrm{d}\theta = C^2\int \sec\theta\ \mathrm{d}\theta$$

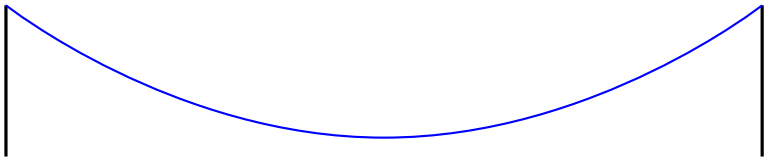
$$= C^2\ln|\sec\theta + \tan\theta| = C^2\ln\left|\frac{\sin\left(\arccos\frac{C}{y}\right) + 1}{\cos\left(\arccos\frac{C}{y}\right)}\right| = C^2\ln|\sqrt{y^2 - C^2} + y| + K$$

which gives the parameterizations

$$\begin{cases} x = \pm\sqrt{y^2 - C^2} + y + K \\ y = e^{\frac{x}{C^2}} \end{cases}$$

$$\begin{cases} x = x \\ y = \frac{1}{2}\left(e^{\frac{x-K}{C^2}} + e^{-\frac{x-K}{C^2}}\right) \end{cases}$$

where the last step is by PlanetMath.



The Catenary

Remark. There is also a simple elementary solution where we set

$$\frac{dy}{dx} = \frac{F_g}{F_t}$$

where F_g is the force of gravity, defined by

$$F_g(x) \propto \int_m^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, \mathrm{d}x$$

where $(m, y(m))$ is a local minimum, and

$$F_t = \alpha$$

is a constant.